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The evolutionary logic of anger and hatred: an empirical test

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ABSTRACT

The emotions of anger and hatred are implicated in both individual and intergroup conflict, aggression, violence, and war. Despite their significance, these emotions are often poorly defined and conflated. Following Sell, Scrivner, Landers, & Lopez, 2021, we provide theoretical groundwork for understanding anger and hatred from an adaptationist perspective, unpacking the evolved functional logic of each emotion that ultimately explains how and why each works the way it does. From this logic, we derive predictions about anger and hatred's behavioral and cognitive outputs and terminating conditions. Using a first person recall design, we then test these predictions across two countries (the United States and United Kingdom) in online participant pools ($n = 725$). In line with theoretically derived predictions, we find striking consistency in the distinctiveness of anger and hatred: Anger produces behaviors functionally designed to bargain for better treatment through recalibration, while hatred produces behaviors functionally designed to neutralize targets costly to the hater. Accurately distinguishing anger from hatred is crucial, as mistaking one for the other can derail conflict resolution and lead to misguided interventions. Viewing anger and hatred through an adaptationist lens not only explains their distinct evolutionary functions but can also guide interventions aimed at reducing conflict at both the individual and societal level.

"Absalom never said a word to Amnon, either good or bad; he hated Amnon because he had disgraced his sister Tamar."

—2 Samuel 13:22 (NIV).

1. Introduction

In nature, actions that benefit the fitness of one organism often diminish the fitness of another. The result is conflict: a powerful selection pressure that over ancestral time has produced adaptations designed to navigate the litany of strategic choices available in aggression and competition, such as whether and how to defer, tolerate, confront, deceive, attack, or destroy. In animals, these choices are computed by neural programs that regulate the output of aggression in different contexts (Archer, 1988; Cosmides & Tooby, 2010; Huntingford & Turner, 1987). Recent evolutionary-psychological work has extended this framework to humans, proposing that the emotions of anger and hatred represent two distinct cognitive programs, each with unique eliciting cues, behavioral outputs, and terminating conditions (Sell et al., 2021; Sell & López, 2020).

Philosophers and psychologists have long debated the relationship between anger and hatred, with some arguing hatred is a generalized form of anger (e.g., Bernier & Dozier, 2002; Frijda, 1986; Power & Dalgleish, 1997), and others distinguishing anger and hatred by their distinct "goals", relationships to moral judgment, or associated appraisal patterns (e.g., Ben-Ze'ev, 1992; Fischer & Roseman, 2007; Halperin, 2011; Halperin, Russell, Dweck, & Gross, 2011; Pretus, Ray, Granot, Cunningham, & Van Bavel, 2023; Smith & Mackie, 2005; Sternberg, 2003, 2005). While much of this work has advanced understanding by focusing on phenomenology (e.g., Lazar, 2003; Steele & Roseman, 2022) or psychological benefits—defending self-esteem, asserting status, and so on (e.g., Averill, 1982; Lazarus, 1991; Lewis, 1971; Lewis, 1987; Miller, 1985; Tangney, Wagner, Fletcher, & Gramzow, 1992), comparatively less research on aggressive emotions has drawn directly on evolutionary principles of adaptive design (Tooby & Cosmides, 1992; Williams, 1966). An adaptationist framework provides a distinctive lens for studying anger and hatred, as it enables us to reverse engineer these

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systems from first principles—asking what adaptive problems they were designed to solve and how natural selection would have structured their features accordingly (Pinker, 2003; Tooby & Cosmides, 1992).

While phenomenological explanations of emotions (e.g., Frijda, 1986; Lazarus, 1991; Mahr & Csibra, 2021; Steele & Roseman, 2022) offer intuitive accounts, they often provide limited predictive power without specifying the underlying adaptive functions and computational structures (Tooby & Cosmides, 1992).

In this paper (see also Sell et al., 2021) we apply an adaptationist framework to anger and hatred, addressing the evolved function of both emotions and their computational architectures.

From an adaptationist lens, it follows that if anger and hatred are two distinct adaptations, they must have been born from two distinct selection pressures and designed to accomplish two distinct functions. Sell et al. (2021) recently proposed what this distinction might be: Anger evolved to bargain for better treatment from those who place too low[1] a weight on the angry person's welfare (i.e., the target of anger, perhaps even a valuable person in one's life, doesn't "care" enough about the angry individual). Indeed, anger typically targets individuals of high cooperative value (often loved ones; Averill, 1983) and works by recalibrating their behavior via threats of cost imposition or benefit withdrawal to restore a more favorable "exchange rate" in the relationship (Sell et al., 2009, 2017). In contrast, hatred evolved to neutralize the net costs imposed by individuals whose very existence and well-being diminishes one's fitness. From the hater's perspective, such "toxic" individuals have by definition negative cooperative value, and hatred functions to depower, distance from, or even kill them (see Sell, 2013; Sell & López, 2020). If indeed these different selection pressures shaped separate adaptations over ancestral time, then anger and hatred should exhibit predictably different cognitive architectures. In what follows, we examine the cognitive maps of anger and hatred individually—their triggers, behavioral outputs, and terminating conditions—with an eye toward understanding precisely how and why these two putatively unique emotions differ.

1.1. Anger as valuation recalibration

To accomplish its bargaining function, the anger system must track changes in how others value the self. Evolutionary psychologists have proposed that such interpersonal valuations are governed by an internal regulatory variable[2] (Tooby, Cosmides, Sell, Lieberman, & Sznycer, 2008) called the *welfare tradeoff ratio* (WTR) which weights another's interests against one's own costs when making decisions (Delton et al., 2023; Lim, 2012; Quillien, 2021; Sell, 2006; Sell et al., 2017). Phenomenologically, WTRs correspond to feelings of "caring" or showing "deference" or "consideration."

Evolutionary researchers have proposed that WTRs come in two functionally distinct types: intrinsic—guiding decisions about others even in their absence—and monitored, adjusting behavior in contexts where one's actions are visible and open to negotiation (Sell, 2006, 2011; Sell et al., 2017; Tooby et al., 2008). Intrinsic WTRs likely evolved to promote behaviors benefiting genetic kin and valuable cooperative partners, as enhancing their fitness often indirectly enhanced one's own (Barclay, 2016; Colnaghi, Santos, Van Lange, & Balliet, 2023; Hamilton, 1963; Lieberman, Tooby, & Cosmides, 2007; Tooby & Cosmides, 1996). Monitored WTRs, by contrast, are sensitive to variables relevant for direct bargaining, such as formidability (one's ability to inflict costs on others; see Delton et al., 2023; Durkee, Goetz, & Lukaszewski, 2018; Hammerstein, 1981; Hammerstein & Parker, 1982; Sell et al., 2009, 2010; Sell, Hone, & Pound, 2012) and cooperative ability (the capacity to bestow or withhold benefits; see Cheng, 2020; Durkee et al., 2018; Landers & Sznycer, 2022; Landers & Sznycer, 2025; Landers, Sznycer, & Al-Shawaf, 2024; Landers, Sznycer, & Durkee, 2024; Sell et al., 2009; Von Reuden, 2014). Thus, anger's functional architecture integrates assessments of kinship, cooperative value, and formidability, among other variables, to recalibrate how much others value one's welfare.

Given this evolutionary backdrop, we argue that anger is best understood as a cognitive mechanism that evolved to bargain for better treatment by recalibrating the welfare tradeoff ratios of others—either by demonstrating one's formidability (ability to inflict costs, Petersen et al., 2013) or cooperative ability (ability to confer benefits) (Sell, 2006, 2011; Sell et al., 2009; Sell et al., 2017). To perform this function effectively, the anger system requires a sophisticated computational structure that can: i) estimate a target's WTR toward oneself, ii) compare that estimated WTR to a negotiated WTR one feels entitled to based on relative bargaining power, kinship, and other relevant variables, and iii) deploy cost-effective strategies, such as threats or actual withdrawal of cooperation or aggression, to raise the target's WTR should it fall below acceptable levels.

1.2. Anger's eliciting conditions and Behavioral strategies

On this bargaining account, anger is triggered primarily when another's actions suggest they undervalue one's welfare. Importantly, anger tracks perceived WTR, not merely inflicted costs. Thus, intentionally harmful actions trigger more anger than unintended harms, and harms directed specifically toward oneself elicit more anger than harms aimed at an anonymous person who happens to be oneself (Sell, 2006; Sell et al., 2017). While direct cost-infliction often triggers anger, even subtle cues of low WTR (e.g., being forgotten or ignored) also reliably evoke anger (Sell, 2006).

Once triggered, anger deploys bargaining tactics to confirm and, if necessary, address a perceived low WTR. To avoid mistakenly assuming undervaluation, anger typically begins with interrogation ("What were you thinking?"). Indeed, rapid communication is anger's most common response (Averill, 1983). Further, research has shown that anger-based arguments follow the "grammar" of WTRs: The angry party highlights the substantial costs they incurred, the minor benefits gained by the offender, and the repeated or intentional nature of the offense (Sell et al., 2017).

If initial arguments fail to resolve the conflict of interest, anger escalates to bargaining directly over the WTR, employing two dominant strategies: cooperative withdrawal (withholding benefits such as communication or intimacy; Williams, Shore & Grahe, 1998) or formidability-based bargaining, leveraging threats or acts of aggression (Enquist & Leimar, 1983; Huntingford & Turner, 1987). Formidability-based bargaining is typically communicative rather than destructive, mirroring patterns in other animals (Enquist & Leimar, 1983; Huntingford & Turner, 1987), progressing from low-cost displays (e.g., staring contests), to moderate cost demonstrations (e.g., pushes and shoves), and then to higher-cost confrontations only if neither opponent yields (e.g., wrestling, punching, weaponry, Sell, Sznycer, & Meyers, 2022). Even lethal aggression typically follows this pattern (Luckenbill, 1977; Felson, 1983), though full-blown physical aggression remains a rare outcome of human anger (Averill, 1982, 1983).

In sum, to recalibrate a target's WTR, anger typically deploys the following strategies:

1. Confront the target to open a channel for bargaining (e.g., "What were you thinking?")
2. Attempt low-cost bargaining in the form of WTR-grammatical arguments (e.g., "you imposed a large cost on me for little benefit")
3. Threaten or withdraw cooperation to leverage cooperative value as a bargaining chip

[1] We expect individual differences in where this threshold is placed (see, e.g., Lukaszewski et al., 2020; Sell, Tooby, & Cosmides, 2009)

[2] Internal regulatory variables are mental constructs that do not fit the common psychological constructs of "belief" or "goal" or "orientation". They are neural variables akin to sensors of hydration levels, that may or may not be conscious but are used by behavioral systems in decision-making.

4. Demand recalibration, e.g., a sincere apology (a signal one has recalibrated and will place more weight on the angry person's welfare going forward), followed by evidence of behavioral change

These strategies are designed to bring about certain behavioral responses in the target. The ultimate goal, in computational terms, is the recalibration of the target's welfare tradeoff ratio, but more proximate goals that enable that recalibration include: 1) the target explaining their behavior (e.g., their perception of the magnitude of the cost imposed so that the target's welfare tradeoff ratio can be better estimated by the angry person), 2) the target listening to the angry person's arguments, 3) the target offering a sincere apology indicating recalibration, and 4) the target demonstrating recalibration by behaving better in the future.

As we will discuss, these strategies and goals are functionally distinct from those of hatred.

1.3. Association value and the adaptive problem of toxic individuals

Anger as a bargaining strategy has two primary limitations. The first is that it is only effective to the extent that one has bargaining power that can be deployed against the target. Therefore, individuals without powerful alliances or status-enhancing traits (e.g., beauty or physical formidability) that improve their bargaining position are less likely to deploy anger (Hess & Hagen, 2006; Sell et al., 2009; Sell et al., 2016). The second is that anger can only limit costs that result from another's low WTR: It cannot prevent others from imposing costs for reasons other than a low WTR—in some cases, an individual's mere presence or success negatively affects one's reproductive success regardless of WTRs. In what follows, we unpack this point from first principles.

The existence and well-being of some individuals impacted our ancestors' welfare and reproductive success to different extents (Aktipis et al., 2018; Tooby & Cosmides, 1996).[3] Some individuals improved one's reproductive success (e.g., family, friends, etc.), while others undermined it (e.g., enemies, those with very low WTRs toward us, rivals, competitors for mates and status). We call this net impact on fitness *association value* (see Petersen et al., 2010, 2012; Sell & López, 2020; Sell et al., 2021). When someone's association value toward you is negative, their existence is "bad" for you. For simplicity, we refer to such individuals as "toxic," though it is important to note that this is a relative term, since what harms one's fitness may benefit another's.

To be clear, association value directly shapes intrinsic welfare tradeoff ratios (i.e., the extent to which you value that individual's welfare intrinsically), much as genetic relatedness (Hamilton's r, Hamilton, 1963) predicts kinship-based valuations (Lieberman et al., 2007). Humans estimate other's association value and embed those estimates in intrinsic WTRs. In the parlance of evolutionary biology, association value is the selection pressure—the state of nature in which another's welfare is causally related to yours—whereas the intrinsic WTR is an internal regulatory variable estimating the association value stored in neural circuitry that calibrates decisions toward that individual.

Thus, negative association values posed an adaptive problem that would have selected for a mechanism dedicated to identifying toxic individuals and deploying cost-effective strategies to neutralize their impact. We propose that hatred is this system (Sell et al., 2021).

If hatred indeed functions to neutralize toxic individuals, its triggers must be ancestrally recurrent cues indicating that another's continued existence or well-being is likely to inflict fitness costs. But how could one possibly know another's association value is negative?

1.4. Triggers of hatred

Based on a functional analysis, Sell et al. (2021) identified four ancestrally recurrent cues to a toxic individual's negative association value:

I. Directly experiencing (on net) large costs from that individual: Past harms reliably predict future threat—if someone has imposed substantial costs on you before, they are likely to do so again (Epstein, 1980; Ouellette & Wood, 1998).
II. Hypothetical reasoning about how one's life would be different if that person did not exist or was diminished in power: Metacognitive abilities allow people to reason counterfactually to estimate the impact of another person on their well-being (e.g., "I would have been better off if you had never been born!").
III. Social learning: Observing reports from trusted others—especially those who share your interests or values—can flag potential threats before personal experience (e.g., learning that Tony betrays friends; see Petersen et al., 2012).
IV. The outputs of other specialized mechanisms that identify individuals with negative association value (e.g., other emotion systems): a conglomerate of specialized mechanisms that can use their computations to identify toxic individuals. For example, jealousy can identify mate poachers and competitors whose continued existence can be detrimental (Buss & Haselton, 2005); disgust can identify individuals who are potential vectors for disease (see Lieberman, Tybur, & Latner, 2012), and so on.

Of course, another predictor that people are toxic to you is their low WTR toward you; after all, such people will happily impose costs on you to gain even trivial benefits. For this reason, anger and hatred should overlap substantially in their eliciting conditions. However, key differences in their elicitors exist as well. For instance, some individuals may genuinely value you yet still undermine your fitness (see Sell, 2012; Sell et al., 2021), such as the woman of your husband's dreams whose mere existence damages your relationship despite them personally holding you in high regard. Anger would be of little use in such circumstances, as recalibrating welfare tradeoff ratios would not solve the problem generated by negative association value.

1.5. Behavioral outputs of hatred

Given anger and hatred's distinct evolved functions, we should also expect to observe fundamentally different behavioral repertoires for each emotion. If hatred indeed evolved to neutralize toxic individuals, its behavioral outputs should reflect strategies for distancing, depowering, or eliminating the threat (Sell et al., 2021).

Distancing removes oneself from toxic individuals, though at the cost of forgoing cooperative opportunities with others who share one's social environment (e.g., avoiding a toxic uncle will prevent you from enjoying the wisdom of your aunt). Attempting to compel the target to distance themselves from you may avoid these costs but requires the power to do so and may be costly to enforce. And some toxic individuals will continue to harm you despite their distance (e.g., the man who stole your girlfriend may stay with her even after one or both of you moves away). Depowering or killing toxic individuals circumvents these limitations but carries risks of its own: potential retaliation and the diminished ability to reestablish a beneficial relationship if one's estimate of their association value was wrong (e.g., Othello discovering his wife was actually loyal after he had already strangled her).

Broadly, all three neutralization tactics—distancing, depowering, and killing—share a common logic of imposing costs onto the toxic target until that target either removes themselves from the hateful

[3] To see this clearly, one can run a hypothetical thought experiment. Imagine an ancient hominid named Eve. At the end of her life, she will have some computable reproductive success—roughly, a count of her proportion of the current human gene pool. If we traveled back in time to her birth and eliminated her neighbor, we could measure the effect of her neighbor's existence on Eve's reproductive success at the end of her life. It will either be positive, negative, or have no effect.

person's social environment, loses status, or dies. To achieve these ends, hatred drives the hateful person's *intrinsic* WTR toward the hated target into negative territory, such that directly imposing costs on the toxic target, even at a cost to the hater, can become a personal benefit. Indeed, the hater may experience the toxic person's suffering as pleasurable (i.e., schadenfreude, see Smith, Powell, Combs, & Schurtz, 2009). However, such spiteful behavior can invite revenge or other reprisals by the hated targets or their allies, leading to risks of increased costs for the hater. For example, as mentioned above, low monitored WTRs invite anger-based bargaining tactics (cost infliction and benefit withdrawal) that may be difficult for the hater to resist. Thus, the hatred system preferentially calibrates *intrinsic* welfare tradeoff ratios, favoring surreptitious cost infliction to minimize reprisals.

Hatred typically remains covert unless circumstances make signaling hatred cost-effective (e.g., everyone in your group also hates the same person or group). Such costly reprisals would have likely been a recurring feature in ancestral environments. For this reason, unlike anger, hatred does not appear to be associated with any facial or behavioral display (e.g., the closed-mouth anger face is known as a universal, species-typical, reliably-developing, coordinated array of muscle contractions that enhance the cues of physical strength in the face as an early bid to demonstrate formidability, see Sell et al., 2014).[4] Instead, hatred often involves withholding information from the target, particularly when its deprivation would be harmful.

In contrast to anger-based aggression, hatred-based aggression employs predatory tactics designed to injure and incapacitate rather than signal formidability; absent are the signals, ritualized displays, and restraints that characterize and constrain anger-based violent conflicts. Rather, hatred may manifest in "unfair" tactics often eschewed in anger-based violence, such as ambushes, surprise attacks, ganging up on an opponent, or waiting until the target is temporarily injured or vulnerable to attack (see Sell et al., 2022). Because the functional product of hatred-based aggression is cost imposition, not recalibration, surrenders and apologies have little to no pacifying effect and may instead provoke further aggression by signaling weakness and vulnerability.

Hatred is also predicted to make use of informational warfare as a low-cost means of damaging a target (see Hess & Hagen, 2006). Because WTRs are calibrated by estimates of traits related to cooperative and formidability-based bargaining, haters can socially depower their targets by spreading malicious information about the target's sexual promiscuity, weakness, cowardice, cooperative deficiencies, or other traits relevant to bargaining power. Even if some or all this information is false, it may have some probability of being believed and thus of lessening the target's social influence (see Petersen, Osmundsen, & Tooby, 2020).

In sum, we propose hatred deploys the following four strategies to neutralize a target:

1. Avoid association with the target
2. Withhold information from the target
3. Deploy information warfare against the target
4. Deploy predatory style aggression against the target

While in anger, avoidance is typically a strategy for recalibration—a threat to withdraw cooperation, hatred typically leads to avoidance of the target on a more permanent basis. And while apologies and conciliatory gestures are useful for signaling a recalibrated WTR in anger, in hatred, apologies and conciliatory gestures may be futile or even counterproductive. Ultimately, the function of hatred is not reconciliation but rather to damage, diminish, or remove the target and their toxicity altogether.

1.6. Predictions: distinct behavioral strategies and goals in anger and hatred

The distinct evolutionary theories of anger and hatred described here require distinct computational designs to carry out their functions: While triggers of anger and hatred will likely overlap to a significant degree, as indicators of low WTR also predict low association value (i.e., a person who trades off your welfare at a low ratio will impose many costs on you, making their existence more costly), anger and hatred's behavioral strategies should be distinct, given that they are designed in response to distinct adaptive problems. In short, anger should trigger behavioral strategies designed to recalibrate the target and bring them back into compliance with a more favorable WTR, while hatred should trigger behavioral strategies designed to neutralize the target by distancing, depowering, or killing the target.

Therefore, we sought to test for differences between these emotions with respect to their different theoretically derived behavioral strategies and goals. The evolutionary models described above predict that participants recalling anger would favor recalibrational strategies and goals (e.g., confrontation, apology), while participants recalling hatred would favor neutralizing strategies and goals (e.g., avoidance, predatory aggression).

2. Methods

We preregistered our study design, analyses, and sample size before data collection began. All data, surveys, preregistrations, code, and supplementary materials can be found in the OSF page: https://osf.io/vpx93/?view_only=e5c1faca6f174b259d0b4d443b720b20. Approval was granted by the social and behavioral sciences institutional review board. Consent was obtained using a standard survey question.

To evoke authentic and ecologically valid manipulations of anger and hatred, we used a between-subjects recall design. After giving consent, each participant was shown a prompt that read: "We are interested in human emotions and would like to ask you some questions about how you felt during these emotions. We need you to think of a person that you felt emotion toward and then answer some questions about how you felt at that time." Participants were then randomly assigned to one of two conditions. Participants in the *anger* condition were asked to think of someone whom they were "very angry at, but do not hate" and asked "Why were you angry with this person?" while participants in the *hatred* condition were asked to think of "the person that you hate most in the world" and asked "Why do you hate this person?" We specifically phrased the anger prompt to avoid eliciting hatred. We presented the prompts this way as an emotion induction (e.g., see Siedlecka & Denson, 2019) and so that the emotion they felt would be at the forefront of their minds when assessing the behavioral items.

After writing about their scenario, participants rated the intensity of their emotion (1 = "not at all" to 7 = "extremely") and then completed items measuring their desire to engage in various behavioral strategies and goals. These items indexed either neutralizing tactics (which we predicted to be preferred in hatred) or recalibrational tactics (which we predicted to be preferred in anger). All hypotheses were tested in two national samples (US, UK).

2.1. Participants

For the US sample, 372 participants were recruited from Prolific. As per the preregistered plan, no participants were removed for failing an attention check, and four participants in the hatred condition were removed for rating their level of hatred as 1 out of 7. No participants in the anger condition rated their level of anger 1. This left a final tally of

[4] An anger face combined with a "smile" will sometimes be perceived as an "evil" face in the same way that someone who expresses pleasure at your suffering will be perceived as an enemy. There is no evidence that such a face is deployed during hatred. Such a face is nowhere to be seen during victim impact statements in the sentencing phase of murder trials, for example.

368 (182 female, 186 male) participants (M_{age} = 38.5, SD = 14.6) in the US sample.

For the UK sample, 369 participants were recruited from Prolific. One participant was removed for failing an attention check. A second participant was removed for rating their anger as 1 in the anger condition, and 10 participants were removed for rating their hatred as 1 in the hatred condition. This left a final tally of 357 (177 female, 180 male) participants (M_{age} = 43.0, SD = 13.7) in the UK sample.

2.2. Measures

After answering the emotion intensity question, participants were presented with 16 behavioral strategies and goals: eight relevant to recalibrating and eight relevant to neutralizing the target of their emotion. Participants rated either how much they felt like taking each action (eight questions) or how much they wanted the target to take each action (eight questions). Each block of eight was presented in random order, with each item presented randomly within each block. All items were rated from 1 (definitely not) to 7 (definitely). See Table 1 for a summary of these items. To diminish self-presentation effects, participants were encouraged to respond with how they truly felt "regardless of whether you felt it was the right thing to do or whether you actually did them. We recognize that people sometimes feel conflicting and opposed desires."

As preregistered, participants' responses on all eight recalibrating strategy and goal items were z-scored and combined to form a summary "recalibrating behavior" variable. We also z-scored and combined all eight neutralizing strategy and goal items to form a summary "neutralizing behavior" variable. This allowed us to test, generally, whether recalibrating strategies and goals are tied to anger, and neutralizing strategies and goals are tied to hatred. Cronbach's alpha for both summary variables were high, indicating strong reliability ($\alpha_{Recalibrating}$: US = 0.83, UK = 0.80; $\alpha_{Neutralizing}$: US = 0.91, UK = 0.88).

After participants answered questions pertaining to recalibrating and neutralizing strategies and goals, we asked participants seven additional questions to further test predicted distinctions between anger and hatred. Whereas our primary items test the core functional differences of anger and hatred via recalibrational and neutralizing strategies, these additional "distinguishing" questions probe related social judgments predicted by our theory. For example, if anger is designed to restore cooperation, angry participants should view apologies as effective, remain willing to listen, and even help a contrite target, whereas hateful participants should not. Likewise, hatred should predict irrational avoidance and unwillingness to reconnect with the target, reflecting a desire to sever toxic ties entirely. By including measures such as willingness to cooperate, belief in a target's capacity for recalibration, and rash avoidance, we can triangulate whether the emotion-specific patterns we see on the 16 strategy/goal items extend to these broader social attitudes. Each of these distinguishing questions was rated from 1 to 7 with specified labels. See Table 2 for a complete list of these questions.

Table 1
Measures of Recalibration and Neutralizing Strategies and Goals.

Recalibration Strategies "Please indicate how much you felt like taking these actions"	Neutralizing Strategies "Please indicate how much you felt like taking these actions"	Recalibration Goals "Please indicate how much you wanted them to take these actions"	Neutralizing Goals "Please indicate how much you wanted them to take these actions"
Confront them to talk about the problem	Wait until they are vulnerable and hurt them	Apologize to you in private	Suffer physically
Explain to them, in private, why they are wrong	Fantasize about killing them	Explain their point of view	Suffer financially
Ignore them for a while so they know what their life is like without you	Never speak to them for the rest of my life	Listen to your point of view	Die
Ask them to apologize so we can put it behind us	Humiliate them in public	Behave better in the future	Leave you alone forever

3. Results

H1. : Did participants in the anger condition prefer recalibrating strategies and goals more than participants in the hatred condition?

Yes. In both samples, an independent samples t-test revealed that participants in the anger condition (US: M = 0.29, SD = 0.50; UK: M = 0.20, SD = 0.55) rated recalibrating strategies and goals higher than those in the hatred condition (US: M = −0.29, SD = 0.70; UK: M = −0.20, SD = 0.67), US: $t(332.66) = 9.07$, $p < .001$, Hedges' $g = 0.94$, 95 % CI [0.73, 1.16], UK: $t(340.19) = 6.15$, $p < .001$, $g = 0.65$, 95 % CI [0.44, 0.86].

H2: Did participants in the hatred condition prefer neutralizing strategies more than participants in the anger condition?

Yes. In both samples, independent samples t-tests revealed that participants in the hatred condition (US: M = 0.50, SD = 0.75; UK: M = 0.41, SD = 0.70) rated neutralizing strategies and goals higher than those in the anger condition (US: M = −0.50, SD = 0.40; UK: M = −0.41, SD = 0.54), US: $t(279.41) = -16.07$, $p < .001$, $g = -1.67$, 95 % CI [−1.90, −1.43], UK: $t(329.93) = -12.47$, $p < .001$, $g = -1.32$, 95 % CI [−1.55, −1.09]. For a summary of these results, see Fig. 1.

3.1. Strategies and goals

As preregistered, the next set of hypotheses distinguishes between

Table 2
The seven "distinguishing" items we administered along with our a priori predictions for the condition that should show a higher score on that item.

Item	Question and Scale	Prediction
Effectiveness of Apologies	"If this person apologized to you sincerely, would you be friends with this person?" (1 = No, definitely not; 7 = Yes, definitely)	Anger > Hatred
Willingness to Cooperate	"If this person lost their job and was in distress, and you knew your company was hiring, would you tell them?" (1 = Yes, definitely; 7 = No, definitely not)	Hatred > Anger
Willingness to Listen	"If this person approached you and asked if you would give them a chance to explain what they did, how willing would you be to hear them out?" (1 = Extremely unwilling; 7 = Extremely willing)	Anger > Hatred
Desire to Communicate	"If the person asked you to explain your point of view, would you want to talk to them?" (1 = No, definitely not; 7 = Yes, definitely)	Anger > Hatred
Recalibratability	"If you were able to explain to the person all the circumstances of their behavior, how likely do you think they would be to realize that they were wrong?" (1 = Extremely unlikely; 7 = Extremely likely)	Anger > Hatred
Intrinsic WTR	"If you found out that this person won the lottery and is getting $250,000, how happy would you be?" (1 = Extremely unhappy; 7 = Extremely happy)	Anger > Hatred
"Irrational" Avoidance	"If you found out that this person owned your car before you did, would you feel like selling it?" (1 = No, definitely not; 7 = Yes, definitely)	Hatred > Anger

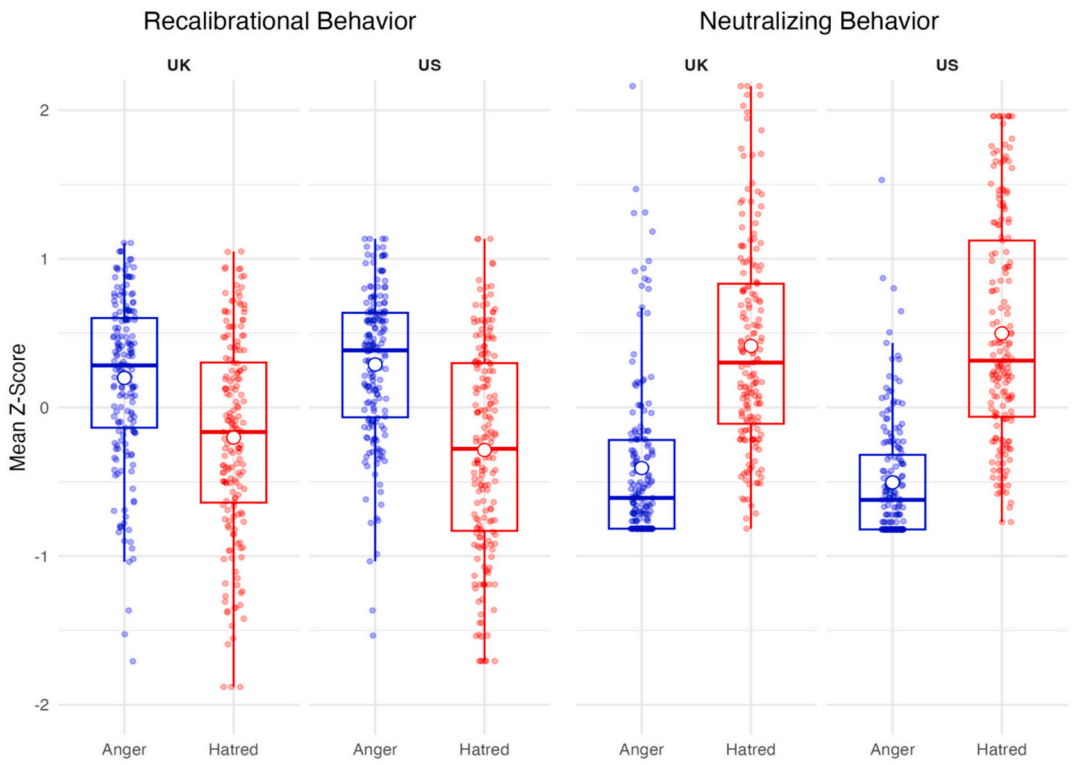


Fig. 1. Distribution of recalibrational and neutralizing behavior scores (z-scored composites) by condition and country. In each panel, the box spans the interquartile range, the thick horizontal line indicates the median, whiskers show non-outlier extremes, and the white circle marks the mean. Individual data points are jittered for transparency.

strategies and goals for both anger and hatred. We took the four questions that correspond to each variable type (strategies & goals, recalibrational or neutralizing) in Table 3 and used them to compute four summary variables. We then tested four hypotheses (H3 - H6): whether recalibrational strategies and goals are preferred in anger or hatred conditions, respectively. Specifically, H3 and H4 examine strategies, while H5 and H6 examine goals.

H3. : Did participants in the anger condition prefer recalibrational strategies more than participants in the hatred condition?

Yes. In both samples, participants in the anger condition (US: $M = 0.26$, $SD = 0.60$; UK: $M = 0.19$, $SD = 0.63$) rated recalibrational strategies higher than participants in the hatred condition (US: $M = -0.25$, $SD = 0.74$; UK: $M = -0.19$, $SD = 0.72$), US: $t(352.95) = 7.27$, $p < .001$, $g = 0.76$, 95 % CI [0.54, 0.97], UK: $t(347.13) = 5.35$, $p < .001$, $g = 0.76$, 95 % CI [0.54, 0.97].

H4. : Did participants in the hatred condition prefer neutralizing strategies more than participants in the anger condition?

Yes. In both samples, participants in the hatred condition (US: $M = 0.44$, $SD = 0.81$; UK: $M = 0.38$, $SD = 0.76$) rated neutralizing strategies higher than participants in the anger condition (US: $M = -0.45$, $SD = 0.45$; UK: $M = -0.37$, $SD = 0.58$), US: $t(286.46) = -13.09$, $g = -1.36$, 95 % CI [−1.58, −1.13], $p < .001$, UK: $t(329.79) = -10.56$, $p < .001$, $g = -1.12$, 95 % CI [−1.34, −0.89].

H5. : Did participants in the anger condition prefer recalibrational goals more than participants in the hatred condition?

Yes. In both samples, participants in the anger condition rated recalibrational goals (US: $M = 0.32$, $SD = 0.53$; UK: $M = 0.21$, $SD = 0.59$) higher than participants in the hatred condition (US: $M = -0.32$, $SD = 0.78$; UK: $M = -0.21$, $SD = 0.82$), US: $t(324.95) = 9.26$, $p < .001$, $g = 0.96$, 95 % CI [0.75, 1.18], UK: $t(321.27) = 5.54$, $p < .001$, $g = 0.59$, 95 % CI [0.37, 0.80].

H6. : Did participants in the hatred condition prefer neutralizing goals more than participants in the anger condition?

Yes. In both samples, participants in the hatred condition (US: $M = 0.56$, $SD = 0.80$; UK: $M = 0.45$, $SD = 0.78$) rated neutralizing goals higher than participants in the anger condition (US: $M = -0.56$, $SD = 0.42$; UK: $M = -0.44$, $SD = 0.56$), US: $t(277.78) = -16.80$, $p < .001$, $g = -1.74$, 95 % CI [−1.98, −1.50], UK: $t(319.82) = -12.36$, $p < .001$, $g = -1.31$, 95 % CI [−1.54, −1.08].

Overall, recalibrational strategies and goals were rated more highly by participants in the *anger condition*, and neutralizing strategies and goals were rated more highly by participants in the hatred condition across both countries. For a summary of these results, see Fig. 2.

3.2. Individual items analysis

As preregistered, independent samples *t*-tests were conducted for each behavioral strategy and goal between conditions for each country.

In the US, each of the 16 individual items (see Table 3) was analyzed for differences between conditions. Almost all results showed all items were significantly different in the predicted direction ($p < .01$, with angry participants preferring recalibrational strategies and goals more than hateful participants, and hateful participants preferring neutralizing strategies and goals more than angry participants). There was one exception, however: There was no significant difference between participants on "ignore them for a while so they know what their life is like without you," $t(364.89) = 0.28$, $p = .783$.

In the UK, each of the 16 individual items (see Table 3) was analyzed for differences between conditions. All items were significantly different ($p < .01$) between conditions in the predicted direction except "ignore them for a while…" (as above), $t(354.03) = -0.82$, $p = .415$ and "behave better in the future" (recalibrational goal), $t(334.58) = 1.10$, $p = .271$. In total, 15 of 16 items showed significant differences in the predicted direction for US participants and 14 of 16 for UK participants. Although

Table 3
Independent-samples t-tests comparing mean endorsement of each behavioral strategy and goal between the Anger and Hatred conditions in the UK and US samples.

Item	UK	US
Recalibrational Strategies	*t*(df), *p*, *g* (95 % CI)	*t*(df), *p*, *g* (95 % CI)
Confront to talk	$t(339.79) = 5.18$, $p < .001$, $g = 0.55$ (0.34, 0.76)	$t(353.28) = 7.52$, $p < .001$, $g = 0.78$ (0.57, 0.99)
Ignore briefly	$t(354.03) = -0.82$, $p = .415$, $g = -0.09$ (−0.29, 0.12)	$t(364.89) = 0.28$, $p = .783$, $g = 0.03$ (−0.18, 0.23)
Explain in private	$t(343.57) = 5.11$, $p < .001$, $g = 0.54$ (0.33, 0.75)	$t(353.84) = 6.57$, $p < .001$, $g = 0.68$ (0.47, 0.89)
Ask for apology	$t(354.81) = 5.60$, $p < .001$, $g = 0.59$ (0.38, 0.80)	$t(365.60) = 6.41$, $p < .001$, $g = 0.67$ (0.46, 0.88)
Recalibrational Goals		
Apologize in private	$t(333.21) = 5.46$, $p < .001$, $g = 0.58$ (0.37, 0.79)	$t(342.91) = 7.66$, $p < .001$, $d = 0.80$ (0.58, 1.01)
Explain their point of view	$t(351.32) = 4.47$, $p < .001$, $g = 0.47$ (0.26, 0.68)	$t(365.23) = 7.34$, $p < .001$, $g = 0.76$ (0.55, 0.98)
Listen to point of view	$t(300.03) = 5.24$, $p < .001$, $g = 0.56$ (0.34, 0.77)	$t(298.77) = 7.99$, $p < .001$, $g = 0.83$ (0.62, 1.04)
Behave better in future	$t(334.58) = 1.10$, $p = .271$, $g = 0.12$ (−0.09, 0.32)	$t(330.50) = 3.24$, $p = .001$, $g = 0.34$ (0.13, 0.54)
Neutralizing Strategies		
Wait until vulnerable→hurt	$t(307.02) = -6.44$, $p < .001$, $g = -0.68$ (−0.90, −0.47)	$t(286.94) = -6.85$, $p < .001$, $g = -0.71$ (−0.92, −0.50)
Fantasize about killing	$t(266.66) = -4.71$, $p < .001$, $g = -0.50$ (−0.71, −0.29)	$t(215.25) = -7.54$, $p < .001$, $g = -0.78$ (−0.99, −0.57)
Never speak again	$t(352.65) = -12.46$, $p < .001$, $g = -1.32$ (−1.54, −1.09)	$t(350.77) = -15.77$, $p < .001$, $g = -1.64$ (−1.88, −1.41)
Humiliate in public	$t(318.76) = -7.94$, $p < .001$, $g = -0.84$ (−1.06, −0.62)	$t(301.19) = -9.51$, $p < .001$, $g = -0.99$ (−1.20, −0.77)
Neutralizing Goals		
Physical suffering	$t(301.96) = -6.85$, $p < .001$, $g = -0.73$ (−0.94, −0.51)	$t(268.99) = -9.81$, $p < .001$, $g = -1.02$ (−1.23, −0.80)
Financial suffering	$t(318.12) = -9.87$, $p < .001$, $g = -1.05$ (−1.27, −0.82)	$t(304.16) = -13.67$, $p < .001$, $g = -1.42$ (−1.65, −1.19)
Die	$t(252.70) = -6.86$, $p < .001$, $g = -0.73$ (−0.94, −0.51)	$t(208.00) = -11.93$, $p < .001$, $g = -1.24$ (−1.46, −1.01)
Leave forever	$t(348.28) = -15.32$, $p < .001$, $g = -1.62$ (−1.86, −1.38)	$t(349.27) = -17.30$, $p < .001$, $g = -1.80$ (−2.04, −1.56)

Note. For each item, the table reports the t-value (with Welch-corrected degrees of freedom), two-tailed *p*-value, Hedges' *g*, and its 95 % confidence interval. Positive *g* values indicate higher endorsement in the Anger condition; negative *g* values indicate higher endorsement in the Hatred condition.

the preregistration specified one-tailed tests for directional predictions, we report two-tailed Welch tests to provide a more conservative estimate; all effects remained significant and in the predicted direction (one-tailed p = two-tailed $p/2$).

3.3. Exploratory analyses

Although our main hypotheses focused on categorical differences between Anger and Hatred conditions, we also explored whether the *strength* of the emotion participants reported systematically predicted their endorsement of recalibrational versus neutralizing strategies and goals.

To do this, we combined each participant's anger-or-hatred rating into a single continuous "Intensity" variable, z-scored it, and ran separate linear models predicting each of our four composite outcomes—Recalibrational Strategies, Recalibrational Goals, Neutralizing Strategies, and Neutralizing Goals—from Intensity, Condition (Anger vs. Hatred), and their interaction, for each country (US vs. UK).

Across each country, the interaction between Intensity and Condition was significant for all four composites ($p < .001$), such that greater intensity of emotion boosted recalibrational strategies and goals only in the anger condition and boosted neutralizing strategies and goals much more so in the hatred condition. For a summary of these interactions across both countries, see Table 4.

We followed up on these significant interactions by analyzing the simple slopes of standardized Intensity by Condition. As expected, greater anger intensity was associated with greater endorsement of recalibrational strategies ($b = 0.22$, 95 % CI [0.12, 0.32], $t(181) = 4.55$, $p < .001$ in the US; $b = 0.23$, 95 % CI [0.13, 0.33], $t(178) = 4.33$, $p < .001$ in the UK) and recalibrational goals ($b = 0.19$, 95 % CI [0.10, 0.28], $t(181) = 4.37$, $p < .001$ in the US; $b = 0.15$, 95 % CI [0.05, 0.25], $t(178) = 2.98$, $p < .001$ in the UK) in both countries, while greater hatred intensity was associated with greater endorsement of neutralizing strategies ($b = 0.31$, 95 % CI [0.22, 0.40], $t(183) = 6.47$, $p < .001$ in the US; $b = 0.35$, 95 % CI [0.26, 0.44], $t(175) = 8.07$, $p < .001$ and in the UK) and goals ($b = 0.37$, 95 % CI [0.28, 0.46], $t(183) = 8.23$, $p < .001$ in the US; $b = 0.33$, 95 % CI [0.25, 0.41], $t(175) = 7.69$, $p < .001$ in the UK). See Fig. 3a-3b and 4a-4b.

Greater anger was modestly associated with increased endorsement of neutralizing strategies ($b = 0.10$, 95 % CI [0.03, 0.17], $t(181) = 2.78$, $p = .006$ in the US; $b = 0.23$, 95 % CI [0.13, 0.33], $t(178) = 4.77$, $p < .001$ in the UK) and goals ($b = 0.12$, 95 % CI [0.05, 0.19], $t(181) = 3.33$, $p = .001$ in the US; $b = 0.17$, 95 % CI [0.08, 0.26], $t(178) = 3.54$, $p < .001$ in the UK) though much less so than recalibrational strategies and goals (see Table 4). Interestingly, greater intensity of hatred was negatively associated with recalibrational strategies ($b = -0.13$, 95 % CI [−0.23, −0.03], $t(183) = -2.69$, $p = .008$) and goals ($b = -0.11$, 95 % CI [−0.21, −0.01], $t(183) = -2.14$, $p = .034$) in the US but not reliably in the UK: (strategies: $b = -0.04$, 95 % CI [−0.13, 0.05], $t(175) = -0.84$, $p = .401$; goals:) $b = -0.07$, 95 % CI [−0.17, 0.03], $t(175) = -1.33$, $p = .185$ in the UK). See Figs. 3a-3b and 4a-4b.

3.4. Distinguishing questions

As part of our preregistered plan, we conducted independent-samples *t*-tests to assess our a priori predictions about how anger versus hatred would influence each of the seven "distinguishing" items. Specifically, we compared participants' mean ratings in the Anger and Hatred conditions on (1) the perceived effectiveness of apologies, (2) willingness to listen, (3) intrinsic WTR, (4) "irrational" avoidance, (5) perceived recalibratability, (6) desire to communicate, and (7) willingness to cooperate (See Table 2).

Across both U.K. and U.S. samples, every comparison was in the predicted direction and showed highly significant differences (all $p < .001$). Angry participants consistently scored higher on measures reflecting openness to reconciliation (effectiveness of apologies, willingness to listen, perceived recalibratability, desire to communicate, and willingness to cooperate) and even reported slightly positive intrinsic WTRs, whereas hateful participants scored higher on measures indicating rejection or avoidance (negative intrinsic WTR and "irrational" avoidance). See Table 5a and 5b for a summary of these results.

4. General discussion

This research provides the first empirical support for the evolutionary account put forth by Sell et al., 2021 that anger and hatred are distinct functional emotions designed for bargaining over welfare tradeoff ratios (WTRs) and reducing the costs of low association value,

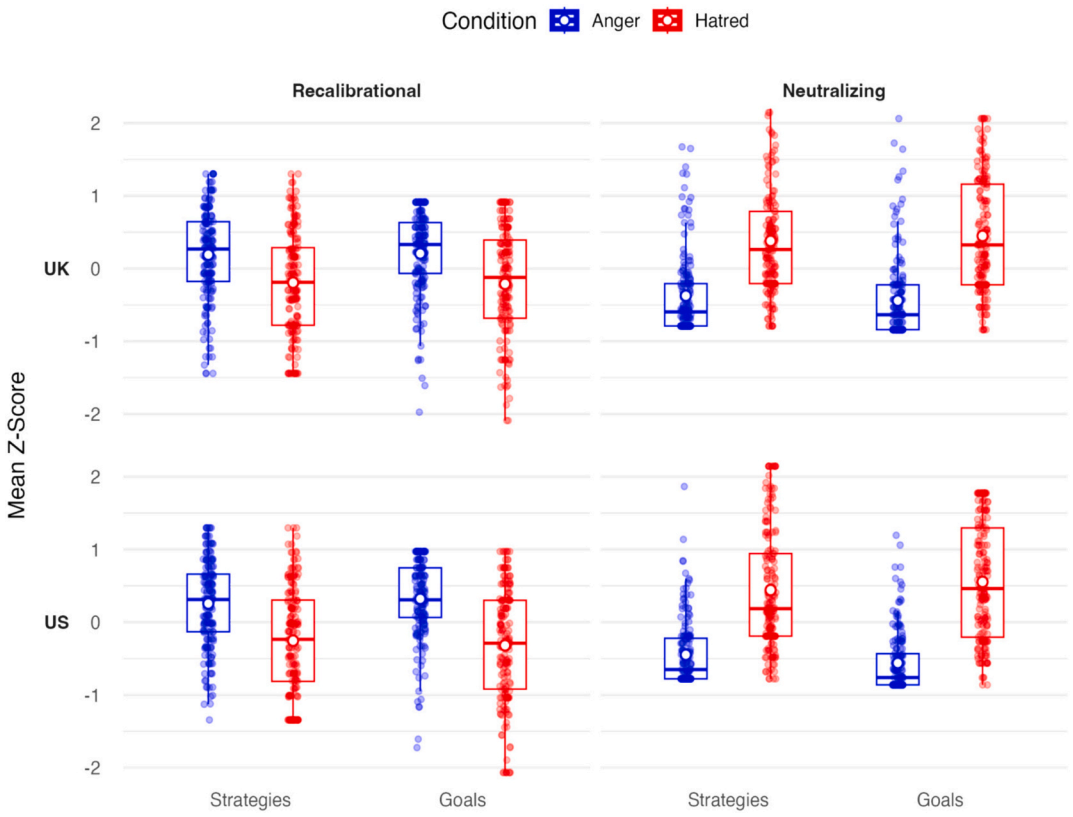


Fig. 2. Distribution of recalibrational and neutralizing strategies and goals (z-scored composites) by condition and country. In each panel, the box spans the interquartile range, the thick horizontal line indicates the median, whiskers show non-outlier extremes, and the white circle marks the mean. Individual data points are jittered for transparency.

Table 4
Emotional Intensity x Condition interactions predicting recalibrational and neutralizing strategies and goals across the US and UK.

Country	Composite	*b*	95 %CI	*t*	*p*
U.K.	Recalibrational Strategies	−0.24	[−0.36, −0.11]	−3.68	< 0.001
	Recalibrational Goals	−0.20	[−0.33, −0.06]	−2.88	0.004
	Neutralizing Strategies	0.10	[−0.01, 0.22]	1.76	0.079
	Neutralizing Goals	0.17	[0.06, 0.29]	2.96	0.003
U.S.	Recalibrational Strategies	−0.30	[−0.42, −0.18]	−4.97	< 0.001
	Recalibrational Goals	−0.26	[−0.38, −0.14]	−4.23	< 0.001
	Neutralizing Strategies	0.18	[0.07, 0.29]	3.15	0.002
	Neutralizing Goals	0.22	[0.12, 0.32]	4.15	< 0.001

Note. Models were estimated separately within each country. Outcomes are composites of within-country z-scored items (higher scores indicate greater endorsement of the named strategy/goal). Intensity is the self-rated emotion intensity (1–7; not standardized). Condition was effect-coded Anger = −0.5, Hatred = +0.5. Reported *b* is the Intensity × Condition interaction and equals the difference in slopes (Hatred − Anger), expressed in *SD* units of the outcome per one-point increase in Intensity. Negative *b* means intensity strengthens recalibrational responding more in the Anger condition; positive *b* means intensity strengthens neutralizing responding more in the Hatred condition. *p* values are two-tailed.

respectively. Across two independent samples (the US and UK), results clearly showed that anger preferentially activated recalibrational strategies and goals, while hatred was consistently associated with neutralizing strategies and goals. Participants experiencing anger were more inclined to engage in dialogue, explain grievances, and seek apologies—behaviors designed to recalibrate others' WTRs. They also scored higher on measures of apology effectiveness, willingness to listen, perceived recalibratability, and desire to communicate. Conversely, hatred not only drove desires to inflict financial, social, and physical harm but also predicted stronger "irrational" avoidance, negative intrinsic WTRs toward the target, and less willingness to cooperate with them—findings that align with the view that hatred evolved to mitigate the negative fitness impact of individuals deemed to have low or negative association value.

Intriguingly, our exploratory analyses provide novel insights into how anger might transition toward hatred. The data show that increased anger intensity was associated not only with enhanced support for recalibrational strategies but also with increased endorsement of neutralizing strategies typically associated with hatred. As expected, anger intensity and its associated tactics escalate when recalibration fails. However, repeated failures to recalibrate a target's WTR could signal that target's persistently low or negative association value, eventually activating hatred to neutralize ongoing costs and prevent future wasted attempts at recalibration. Consistent with this account, greater anger intensity moderately predicted increased support for neutralizing strategies (though less strongly than hatred), whereas greater hatred intensity predicted diminished support for recalibrational strategies in the US and directionally so in the UK. Pooled across countries, this association was small but reliable ($\beta \approx -0.09$, 95 % CI [−0.15, −0.03]), indicating that greater hatred intensity tends to coincide with reduced support for recalibration. Given that the UK estimate trended in the same direction but was not individually reliable, this cross-country pattern should be replicated with larger samples.

Notably, although our results almost unanimously fell in line with our predictions, one measure consistently failed to differentiate anger from hatred across countries: "Ignore them for a while so they know what their life is like without you." One possibility we considered is that

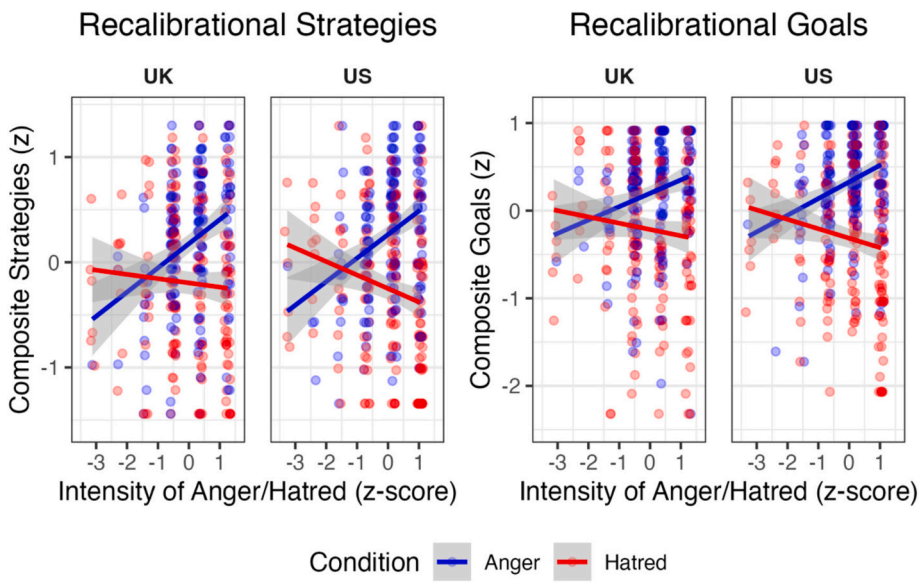


Fig. 3. a-3b: Intensity of Anger/Hatred (z-score within country) predicting Recalibrational Strategies (3a) and Recalibrational Goals (3b), shown separately for the UK (left panels) and US (right panels). Outcome composites were created by z-scoring each item within country and then averaging (higher values = greater recalibrational endorsement). Solid lines are separate ordinary least squares regression fits by condition with shaded 95 % confidence bands; points are jittered for visibility.

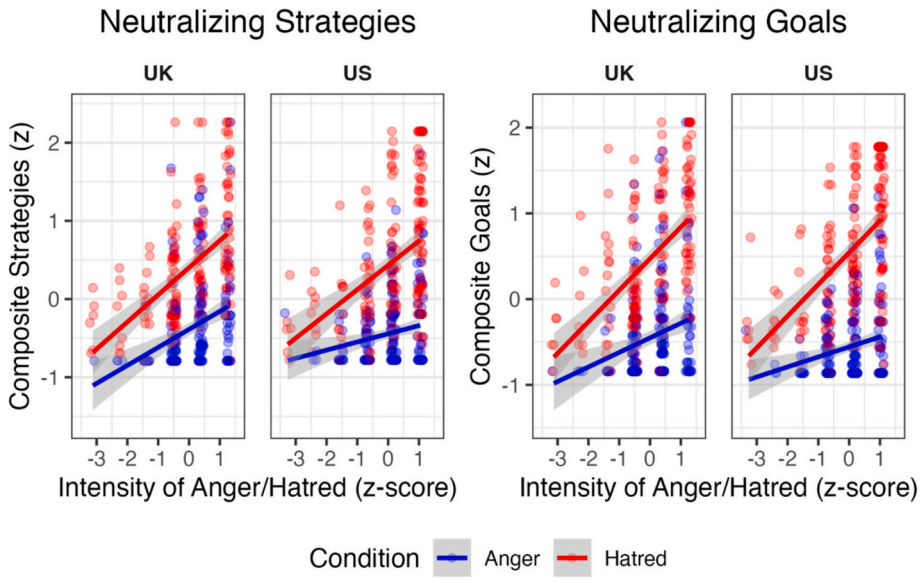


Fig. 4. a-4b: Intensity of Anger/Hatred (z-score within country) predicting Neutralizing Strategies (4a) and Neutralizing Goals (4b), shown separately for the UK (left panels) and US (right panels). Outcome composites were created by z-scoring each item within country and then averaging (higher values = greater neutralizing endorsement). Solid lines are separate ordinary least squares regression fits by condition with shaded 95 % confidence bands; points are jittered for visibility.

participants in the Hatred condition may have interpreted 'ignore them for a while' as permanent rather than short-term avoidance. The data show mixed evidence for this: Among participants in the Hatred condition, 'ignore them for a while' correlated only weakly with our two permanent-avoidance measures in the US (leave forever: $r = 0.08$; never speak: $r = 0.12$ in the US) but more strongly with the other recalibrational items (strategies: r's = 0.13–0.25; goals: r's = 0.19–0.35). However, this pattern reversed in the UK: Correlations with permanent avoidance were larger (leave forever: $r = 0.19$, never speak: $r = 0.34$) than correlations with the other recalibrational items (strategies: r's = 0.07–0.10; goals: r's = 0.06–0.12). This pattern suggests that, at least in the UK, some participants may have misinterpreted 'ignore them for a while' as long-term avoidance, making it less clearly a recalibrational item than the rest. Furthermore, we speculate that low 'ignore them for a while' scores in the Anger condition may reflect a second-order anger response—an escalatory strategy reserved for times when first-line, direct recalibration (confrontation, explanation, obtaining an apology) has failed (see Sell, 2011). In other words, because anger's core function is to restore acceptable levels of WTR, and those levels can be restored most cost-effectively through direct engagement, participants may have read support for the "silent treatment" as a tacit endorsement of and evidence for those initial bargaining attempts failing. Thus, rather than indicating a theoretical misalignment, the muted Anger–Hatred divergence on this item may underscore the sequential, escalation-based nature of anger-driven recalibration.

In addition to evaluating new theoretical predictions about anger

Table 5a
U.K. sample: Descriptive statistics and comparisons for distinguishing items.

Item	Anger *M* (*SD*)	Hatred *M* (*SD*)	*t*(df)	*p*	*g*
Effectiveness of Apologies	0.69 (0.77)	−0.71 (0.66)	*t*(348.20) = 18.51	<0.001	1.95
Willingness to Listen	0.49 (0.71)	−0.50 (1.01)	*t*(315.53) = 10.79	<0.001	1.14
Intrinsic WTR	0.57 (0.96)	−0.58 (0.64)	*t*(314.00) = 13.44	<0.001	1.42
Irrational Avoidance	−0.49 (0.99)	0.50 (0.73)	*t*(330.17) = −10.68	<0.001	−1.12
Recalibratability	0.44 (1.00)	−0.45 (0.78)	*t*(339.10) = 9.43	<0.001	0.99
Desire to Communicate	0.47 (0.72)	−0.48 (1.02)	*t*(316.43) = 10.04	<0.001	1.06
Willingness to Cooperate	−0.38 (0.79)	0.39 (1.04)	*t*(327.99) = −7.83	<0.001	−0.83

Note: Table 5a reports item, mean response and standard deviation in the Anger condition, mean response and standard deviation in the Hatred condition, the t-value (with Welch-corrected degrees of freedom), two-tailed *p*-value, and Hedge's *g* in the UK sample.

Table 5b
U.S. sample: Descriptive statistics and comparisons for distinguishing items.

Item	Anger *M* (*SD*)	Hatred *M* (*SD*)	*t*(df)	*p*	*g*
Effectiveness of Apologies	0.71 (0.69)	−0.70 (0.73)	*t*(365.09) = 18.94	<0.001	1.97
Willingness to Listen	0.60 (0.67)	−0.59 (0.92)	*t*(336.55) = 14.07	<0.001	1.46
Intrinsic WTR	0.64 (0.86)	−0.63 (0.69)	*t*(348.54) = 15.64	<0.001	1.63
Irrational Avoidance	−0.47 (0.95)	0.46 (0.82)	*t*(356.47) = −10.02	<0.001	−1.04
Recalibratability	0.55 (0.91)	−0.54 (0.76)	*t*(352.71) = 12.43	<0.001	1.29
Desire to Communicate	0.56 (0.63)	−0.56 (0.99)	*t*(310.55) = 12.93	<0.001	1.34
Willingness to Cooperate	−0.47 (0.74)	0.47 (1.00)	*t*(339.17) = −10.23	<0.001	−1.06

Note: Table 5b reports item, mean response and standard deviation in the Anger condition, mean response and standard deviation in the Hatred condition, the t-value (with Welch-corrected degrees of freedom), two-tailed p-value, and Hedges' *g* in the US sample.

and hatred (and their divergence), our findings dovetail with and advance alternative functional frameworks of these emotions. Recent functional theories of hatred (e.g., Fischer, Halperin, Canetti, & Jasini, 2018; Van Doorn, 2018) have emphasized appraisal patterns (e.g., stability attributions) and perceptions of negative dispositions (e.g., seeing the target as evil or full of mal intent), suggesting individuals are hated when viewed as intentionally harmful and unchanging threats. Our evolutionary-functional account aligns partially with these views but points to a different elicitor: Neither unstable attributions nor perceptions of unintentional harm may preclude hatred when a target's mere existence consistently inflicts fitness costs (e.g., a socially stigmatized individual who adores you); and neither stable attributions nor perceived intentional harm may necessitate hatred when the target's mere existence consistently provides benefits (e.g., a skilled hunter who despises you but whose kills feed the whole village, including your family). From this perspective, appraisals are not themselves primary elicitors of emotion but rather downstream cognitive concomitants of emotions (in the case of stability) or reflect individual cues (in the case of intentionality) embedded within a broader computation of others' association value. This shift—from treating appraisals as intrapersonal "first movers" to viewing them as computational correlates or pieces within a larger evolved, interpersonal puzzle—opens a new avenue for future research to map how appraisal patterns interact with core interpersonal emotion architectures.

More broadly, our findings speak to a central challenge in emotion science: the need to carve emotions at their functional joints. Emotion researchers have long sought clear theoretical boundaries between emotions, and the present work advances that effort. When we fail to distinguish emotions by their adaptive functions, we risk conflating distinct mechanisms that evolved for different purposes, obscuring their computational designs and misinterpreting their psychological, behavioral, and physiological signatures. Indeed, when researchers rely on overlapping theoretical frameworks—treating, say, anger and disgust or envy and resentment as interchangeable—they risk producing noisy or contradictory findings (Landers & Shaw, 2025; Sznycer, Cosmides, & Tooby, 2017). For instance, despite the copious quantity of ink spilled attempting to reconcile intrapersonal and interpersonal research on shame, relatively little progress has been made (though see Landers, Sznycer, & Durkee, 2024); the same problem plagues research on envy and resentment (see Cohen-Charash & Larson, 2017a, 2017b; Crusius, Gonzalez, Lange, & Cohen-Charash, 2020; Landers & Shaw, 2025) as well as many other emotion literatures. From an evolutionary-functional standpoint, each emotion embodies a distinct computational design tuned to specific fitness pressures; imprecise definitions thus undermine both theoretical clarity and empirical prediction, such that manipulations intended to elicit one emotion may inadvertently recruit another (Landers & Shaw, 2025). We encourage future work to ground emotion constructs in their evolved functions and hope this work serves as a practical roadmap for researchers interested in building and then testing evolutionary-functional theories from first principles.

Moreover, our findings offer practitioners a functional map for distinguishing anger and hatred—an essential step for effective intervention in aggressive conflicts (McDermott, Lopez, & Hatemi, 2017). As an emotion functionally designed for bargaining over treatment, anger's hallmark is openness to dialogue and desire for explanations and apologies. Practitioners who recognize these signatures can focus on facilitating constructive bargaining by creating channels for direct communication, encouraging perspective-taking about the costs and benefits of the offending behavior, and guiding both parties toward tangible behavioral change. By contrast, because hatred reflects the judgment that a target's association value is negative, attempts to encourage reconciliation may largely be futile and may even be counterproductive. Recognizing hatred's behavioral hallmarks—desire to permanently separate from, depower, and/or harm—can help practitioners redirect efforts away from reconciliation and toward safe, nonviolent neutralization strategies, such as restructuring group boundaries or reducing contact. By correctly identifying which emotion is activated, professionals in clinical, organizational, or political arenas can match their interventions to the evolved functions of each emotion, increasing the odds of de-escalation and reducing the risk of misapplied conflict-resolution tactics.

Several limitations of this work should be acknowledged. First, reliance on self-report constrains behavioral validity, despite evidence indicating self-report surveys reliably approximate real-world behaviors (Paulhus & Vazire, 2007; Robins, Hendin, & Trzesniewski, 2001). Future research should incorporate behavioral measures or ecological momentary assessment to validate these findings. Second, the fact that we sampled only from WEIRD populations the (US and UK) limits the generalizability of our findings (Henrich, Heine, & Norenzayan, 2010). Future work should attempt to replicate our results cross-culturally. Investigating non-WEIRD populations could also illuminate whether anger and hatred universally produce recalibrational and neutralizing strategies and goals, respectively, or whether these strategies and goals vary meaningfully across emotional contexts within diverse populations.

4.1. Conclusion

Our analysis provides preliminary evidence that anger and hatred are not simply gradations of the same emotional program but instead

have distinct adaptive functions finely tuned to different social dilemmas: Anger operates as a bargaining mechanism, recalibrating welfare-tradeoff ratios when they fall too low, whereas hatred functions to neutralize targets whose low association value imposes high costs. These functional signatures not only deepen our understanding of emotional complexity but also carry clear implications for both theory and practice. Future work should continue to map emotions onto their evolved functions, ensuring that our theorizing aligns with the specific adaptive challenges each emotion was designed to solve.

CRediT authorship contribution statement

Mitchell Landers: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Validation, Visualization, Writing – original draft, Writing – review & editing. **Aaron Sell:** Conceptualization, Formal analysis, Investigation, Methodology, Project administration, Supervision, Writing – original draft, Writing – review & editing. **Coltan Scrivner:** Conceptualization, Data curation, Formal analysis, Validation, Writing – review & editing. **Anthony Lopez:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Resources, Validation, Visualization, Writing – review & editing.

Declaration of competing interest

None.

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